

Expense Variance & Budget Analysis

500-Person Technology Company · January 2023 – December 2024

55,348 expense transactions across 8 departments and 29 categories — revealing systemic overspend patterns and actionable opportunities for budget discipline.



Executive Summary

\$135.57M

Total Budget (2Y)

\$159.81M

Total Actual Spend

\$24.24M

Total Overspend

17.9%

Overall Variance

73%

Transactions Overspent

Of all 55,348 transactions, 72.59% were classified as Overspend, 22.45% On-Track, and 4.96% Underspend — confirming budget discipline is the exception, not the norm.



Key Findings

Engineering Dominates

\$75.16M budget (55% of total) with a consistent 22.5% overspend across both years.

Three Chronic Overspenders

Engineering, Sales, and Legal all exceed budget by 22%+, driven by salaries, travel, and legal fees.

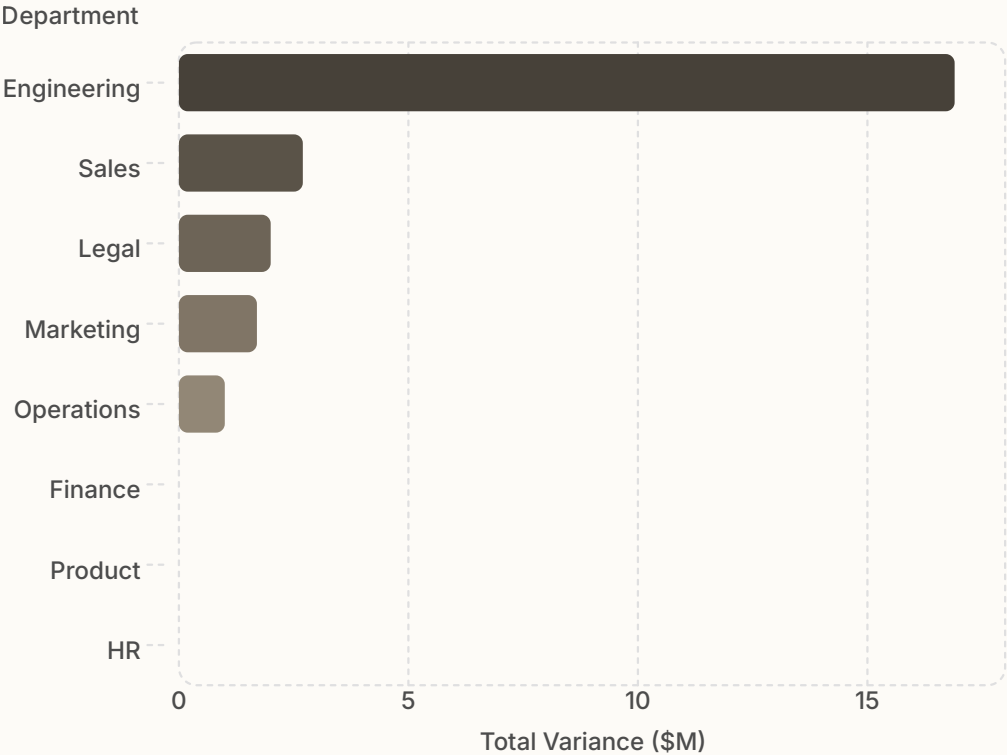
Seasonal Spikes

Marketing and Operations peak in Q2/Q4, with overspends reaching 40% in high-activity months.

Top Variance Drivers

Salaries (\$8.10M) and Cloud Infrastructure (\$4.50M) together represent over 52% of total variance.

Dashboard: Total Variance by Department



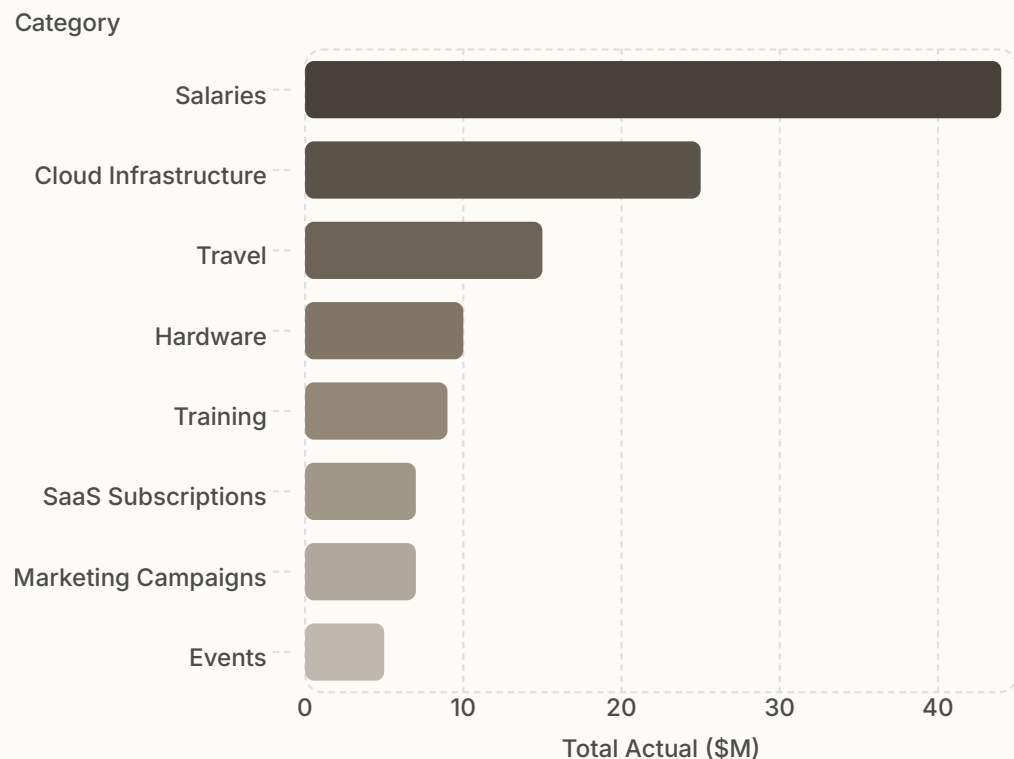
Engineering Leads by a Wide Margin

Engineering's \$16.9M variance is more than 6× the next highest department (Sales at \$2.7M), reflecting structural budget misalignment in the largest cost centre.

Finance, Product, and HR show near-zero variance — benchmarks for budget discipline.

⚠ Engineering's variance alone accounts for nearly 70% of total company overspend.

Transaction Status & Spend by Category



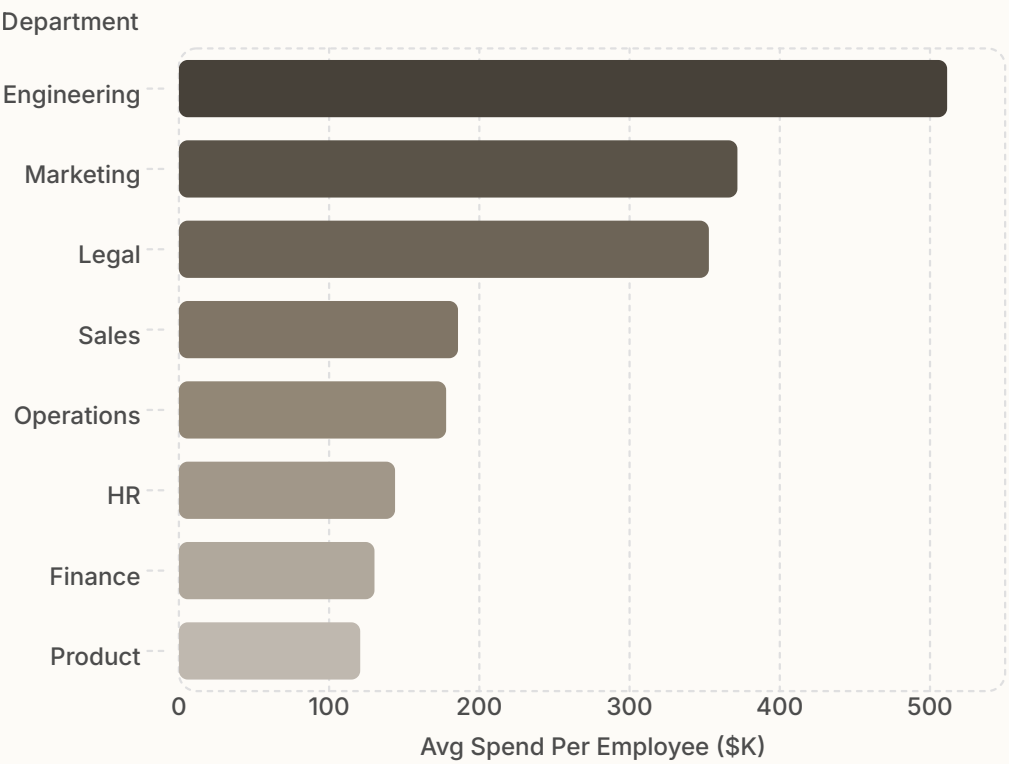
Where the Money Goes

Salaries (\$44M) and Cloud Infrastructure (\$25M) are the two largest categories, together accounting for 43% of total actual spend.

Travel (\$15M), Hardware (\$10M), and Training (\$9M) round out the top 5 — high discretionary spend that may be controllable with tighter policy.

📌 72.59% of all 55K transactions are classified as Overspend — a systemic, not isolated, issue.

Department Performance at a Glance



Spend Per Employee

Engineering's \$511.4K per employee is nearly 45% above the company average. Higher spend per employee directly correlates with increased budget deviation.

Legal: 22.8% Variance

Highest % overspend — unpredictable legal fees and consulting

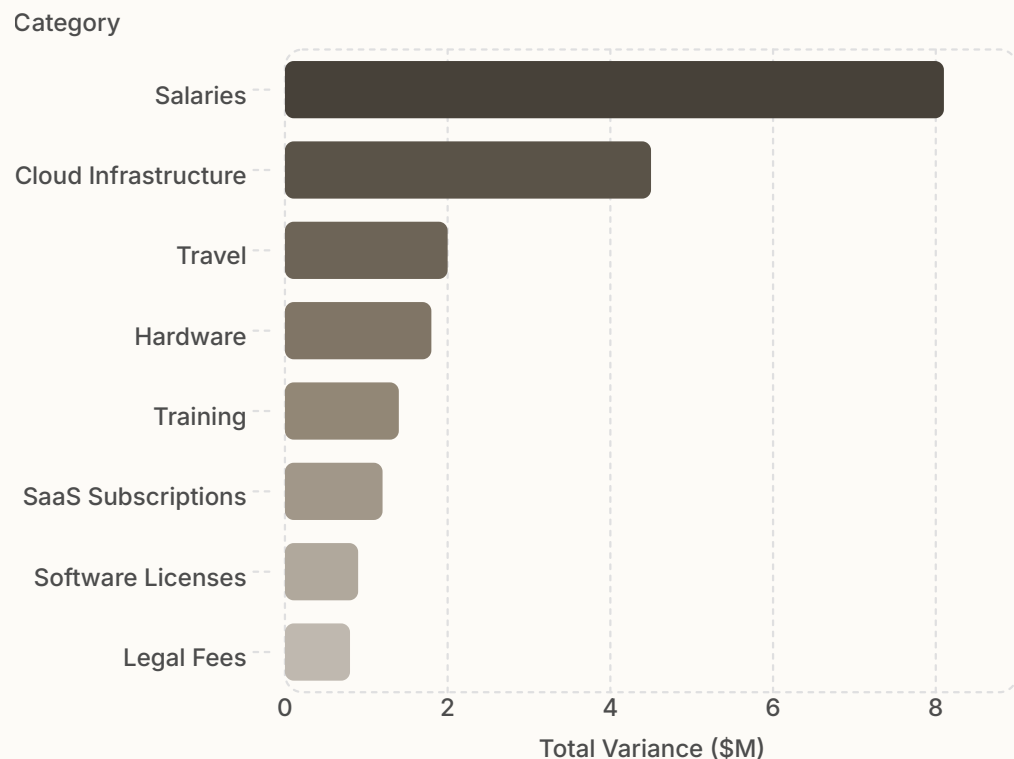
Finance & Product: 99%+ FA

Near-perfect forecast accuracy — the benchmark

HR: -0.4% Variance

Only department underspending — likely unfilled headcount

Expense Risk & Overspend by Category



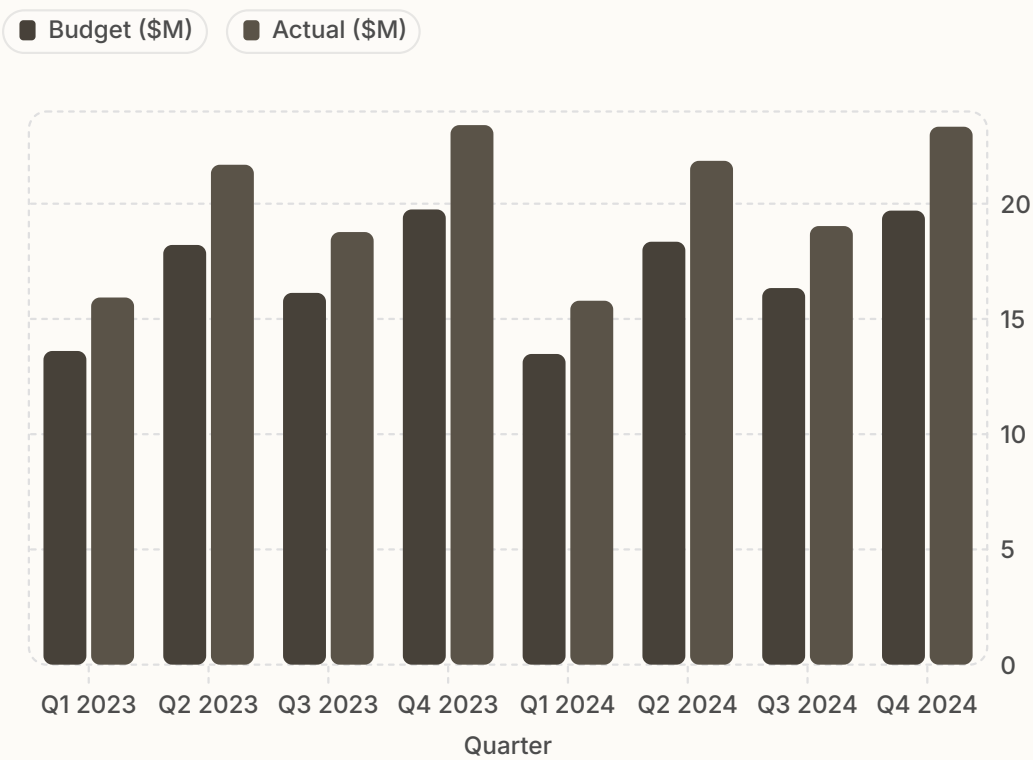
Structural, Not Anomalous

Salaries (\$8.10M) and Cloud Infrastructure (\$4.50M) dominate variance. 15,037 transactions (27% of all) are classified as CriticalOverspend.

All 8 departments are flagged as high-risk — overspending is concentrated in recurring operational categories, requiring policy intervention, not exception management.

⚠ The long-tail of smaller categories shows near-zero variance, confirming well-managed low-volume spend.

Quarterly Performance: Consistent Overspend Pattern



Year-Over-Year: No Improvement

The overspend rate held steady at 17.9% across both years — 2023: \$12.10M variance; 2024: \$12.14M variance. Q2 and Q4 consistently show the highest absolute overspend.

2023 Total

\$67.69M budget → \$79.79M actual (+17.9%)

2024 Total

\$67.88M budget → \$80.02M actual (+17.9%)

Recommendations

01

Hard Budget Caps for Engineering

Monthly CTO reviews; hard caps on cloud spend above \$8,000 per submission.

02

CFO Pre-Approval for High-Risk Categories

Mandatory approval for Salaries, Cloud, and Legal Fees above \$10,000 — 75% of total variance.

03

Replicate Best Practices

Document Finance, HR & Product's budgeting methodology and roll out to Engineering and Sales.

04

Seasonal Rebalancing & Automated Alerts

Front-load Marketing & Operations budgets for Q2/Q4 spikes; trigger alerts at 90% utilisation.

05

Audit Legal Fees & Address Systemic Overspend

Renegotiate vendor contracts; launch company-wide budget awareness initiative with monthly scorecards.

Appendix: Data Dictionary & Sources

Key Fields

expense_id	Unique record identifier (EXP-XXXXX)
variance_status	Overspend / On-Track / Underspend
variance_severity	On-Track / Moderate / Critical Overspend
budget_utilization_pct	Actual / Budget × 100
approver	Manager / Director / VP / CFO

Data Sources & Tools

expense_data_cleaned_final.csv	Primary dataset — 55,348 rows, 22 columns
Expense_Variance__Budget_Analyses.ipynb	Jupyter notebook for EDA and variance analysis
Dashboard.pbix	Power BI file — 3 pages: Overview, Operations, Risk
generate_dataset.py	Python script for synthetic 50K+ row dataset
expense_queries.sql	SQL queries for database exploration and validation

❏ Report generated 22 May 2026. All figures in USD. Data covers January 2023 – December 2024 across 8 departments and 29 expense categories.